Term Paper

On

Loan Default Prediction Using Random Forest and Logistic Regression

Submitted to

Amity University, Uttar Pradesh

In partial fulfilment of the requirements for the award of the degree of

Bachelor of Science (Artificial Intelligence and Data Science)
(Honours/Honours with research)

In

Amity Institute of Information Technology (AIIT)

By

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DECLARATION

I Mannatveer Singh student of BSc. (AI&DS) (H/HR) (2024-2028) hereby declare that the termpaper titled "Loan default prediction using random forest and logistic regression" which is submitted by me to Dr. Neha Prerna Tigga Amity Institute of Information Technology Amity University Uttar Pradesh, Noida, in partial fulfillment of therequirement for the award of the degree of BSc. (AI&DS) (H/HR) has not been previouslyformed the basis for the award of any degree, diploma or other similar title or recognition.

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CERTIFICATE

On the basis of declaration submitted by Mannatveer Singh, student of BSc. (AI&DS) (H/HR), I hereby certify that the Term Paper titled “Loan Default Prediction Using Random Forest and Logistic Regression” which is submitted to Department of Information Technology, Amity Institute of Information Technology, Amity University Noida, in partial fulfilment of requirement for the award of the degree of Bachelor of Science (Artificial Intelligence and Data Science) (Honours/Honours with research) is an original contribution with existing knowledge and faithful record of work carried out by them under my guidance and supervision.

To the best of my knowledge this work has not been submitted in part or full for any Degree or Diploma to this University or elsewhere.

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of project work and its report submission.

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Project Information

1) Project Duration : **(33 Days)**

a) Date of Summer Internship commencement **(02/06/2025)**

a) Date of Summer Internship Completion **(04/07/2025)**

2) Topic

Loan Default Prediction Using Random Forest and Logistic Regression

3) Project Objective

The primary objective of this project is to develop and evaluate predictive models, specifically using Random Forest and Logistic Regression, to accurately forecast the likelihood of loan default. This involves data preprocessing, feature engineering, model training, and rigorous performance assessment to identify the most effective model for minimizing financial risk and improving lending decisions.

4) Methodology to be adopted

1. Data Collection and Preparation 2. Exploratory Data Analysis (EDA) 3. Feature Engineering ~Random Forest ~Logistic Regression 4. Model selection and evaluation 5. Model Deployment and Monitoring

5) **Brief Summery of project(*to be duly certified by the industry guide*)**

This project aims to predict loan defaults using machine learning models. It involves collecting and preparing loan data, exploring the data to find patterns, and creating new features. Two models, Random Forest and Logistic Regression, are trained and evaluated to determine which one is best at predicting whether a loan will default. The project's goal is to create a reliable tool for assessing loan risk.

Signature (Student)	Signature (Industry Guide)	Signature (Faculty Guide)

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ABSTRACT

In this modern digital era, the banking and financial sectors go through growing challenges related to loan default risks. Exact prediction of loan defaults is essential for reducing financial losses and improving credit evaluation. This term paper rectifies the execution of two machine learning methods—**Random Forest** and **Logistic Regression**—for predicting loan default.

Logistic Regression, a statistical method, calculates the percentage of default based on linear relationships among variables. However, Random Forest, an ensemble learning method, captures nonlinear relationships and interactions among numerous features by using a collection of decision trees. By analyzing a publicly available loan dataset, this study creates contrast between the performance of both models in terms of accuracy, precision, recall, and AUC score.

This paper also discusses the ethical inferences of algorithmic decisions in financial services, such as objectivity in lending and transparency of prediction models. The results show that machine learning models can improve risk prediction accuracy, with Random Forest performing better than Logistic Regression in most metrics. Moreover, careful handling of data privacy and bias remains important to deploy such systems responsibly.

INTRODUCTION

Loan default prediction is a key regard for financial firms. When a client fails to payback a loan, it leads to financial loss and dissatisfies lending institutions. Current credit scoring models are often insufficient in identifying defaulters, especially with the availability of immense and difficult data. Machine learning provides powerful tools to improve prediction by identifying unnoticeable patterns in data.

Two major machine learning algorithms used in credit scoring are **Logistic Regression** and **Random Forest**. Logistic Regression is a dependent model useful for binary classification problems, usually when interpretability is a major demand. On the other hand, Random Forest is a non-parametric ensemble model capable of handling high-dimensional and nonlinear data.

This project aims to apply two of the algorithms to a loan dataset, create a contrast between their performance, and showcase which model provides efficient results for default prediction. It also enhances the ethical challenges in automated credit risk assessment, including data bias, transparency, and accountability.

LITERATURE REVIEW

Numerous studies and papers have been held in the field of credit risk prediction using machine learning. The conventional models based on financial ratios and credit scores are now being replaced by machine learning practices.

- **Baesens et al. (2003)** featured the excellence of ensemble models such as Random Forests over logistic models due to their greater predictive power.
- **Lessmann et al. (2015)** contrasted 41 classification algorithms and got to know that Random Forests to be constantly among the top performers for credit scoring.
- **T. Bellotti and J. Crook (2009)** found that logistic regression remains in demand in banks due to its simplicity and ease of interpretation.

While Logistic Regression offers precision and directness, Random Forest gives greater accuracy at the cost of interpretability. Several researchers emphasize the neccessity of creating a balance between performance with ethical considerations like transparency and bias reduction.

Recent developments also suggest that collaboration of machine learning models with explainable AI (XAI) methods can lead to fair and accountable credit assessment tools.

METHODOLOGY

- *DESCRIPTION OF DATASET*

The dataset referred for this indication is a public loan dataset received from **Kaggle**, containing information on loanees such as:

- Amount of loan
- Yearly income
- Credit score
- Purpose of loan
- Terms and conditions of the loan
- Work status
- Loan (Defaulted or Completely Paid)

- *PROCESSING OF DATA*

- Blank values were handled using mean/mode computation.
- Categorical variables were made codes using one-hot encoding.
- Feature scaling was used by referring to standardization for Logistic Regression.
- Class imbalance was taken care of using SMOTE (Synthetic Minority Oversampling Technique).

- *MODEL SELECTION*

- **Logistic Regression** was used as the baseline model due to its extensive use in credit risk evaluation.

- **Random Forest Classifier** was opted for its robustness and its skills to handle complex interactions.

- EVALUATION METRICS

The models were assesed based on:

- Accuracy
- Precision
- Recall
- F1-Score
- ROC-AUC Score

A **10-fold cross-validation** was used to ensure reliable performance estimates

IMPLEMENTATION

- *LOGISTIC REGRESSION*

Executed using **Scikit-learn** library in Python, Logistic Regression was instructed on the preprocessed dataset. The model coefficients provided perceptions into which details (e.g., loan amount, credit score) influenced the default prediction.

- **Advantages**: Simple to implement, interpretable, fast preparation
- **Limitations**: Determines linear relationship, less accurate for complex datasets

- *RANDOM FOREST*

Random Forest was executed using the **RandomForestClassifier** from Scikit-learn. Key hyperparameters like number of trees, maximum depth, and minimum samples split were tuned using **GridSearchCV**.

- **Advantages**: Greater accuracy, handles non-linearity, robust to overfitting
- **Limitations**: Lower interpretable, higher computation cost

- *TOOLS AND TECHNOLOGIES USED*

- Python (Jupyter Notebook)
- Libraries: Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn
- SMOTE for class balancing
- GridSearchCV for hyperparameter tuning

COMPARATIVE ANALYSIS: RANDOM FOREST VS LOGISTIC REGRESSION

Metric	Logistic Regression	Random Forest

Accuracy	83.4%	91.2%
Precision	76.5%	88.7%
Recall	72.3%	90.1%
F1-Score	74.3%	89.4%
ROC-AUC Score	0.81	0.94

- INTERPRATATION

- Random Forest clearly performed better than Logistic Regression in all metrics.
- Logistic Regression, although less accurate, provided clearer interpretability of major factors leading to loan default.
- ROC curves demonstrated Random Forest's better discrimination power between defaulters and non-defaulters.

-VISUALISATION

Graphs such as confusion matrix, ROC curves, and feature importance plots were used to showcase model performance and influential factors.

ETHICAL ANALYSIS

While predictive models offer enhanced efficiency, their deployment in sensitive domains like credit lending gives elevation to ethical concerns:

- *BIAS AND FAIRNESS*

- If previous data is biased (e.g., gender or racial bias), models may perpetuate or diminish these inequalities.
- Equitable lending laws such as **Equal Credit Opportunity Act (ECOA)** need transparency and fairness in decision-making.

- *TRANSPARENCY*

- Logistic Regression is interpretable, while Random Forest can be opaque.
- Explainability tools like **SHAP** and **LIME** should be combined with complex models for accountability.

- *DATA PRIVACY*

- Sensitive information (income, employment) must be kept under regulations like **GDPR** or **India's DPDP Act**.
- Institutions must make sure that encrypted storage and ethical data handling.

- *HUMAN OVERSIGHT*
- AI systems should enhance, not replace, human decision-making.
- Final loan decisions must involve human verification, especially in ambiguous cases.

CONCLUSION

The analysis illustrates the effectiveness of machine learning models in predicting loan default. While Logistic Regression offers transparency and simplicity, Random Forest offers better accuracy and robustness.

For financial institutions, utilizing such systems can reduce default rates significantly and improve lending decisions. However, ethical deployment is important to ensure fairness, accountability, and trust.

Future improvements may involve using **deep learning**, **ensemble stacking**, or **hybrid models** for further performance enhancement. Additionally, integrating **explainable AI (XAI)** frameworks will help balance predictive power with interpretability and ethics.

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